Dual Imbalance Suppression for Source-Free Object Detection in Aerial Images

You Ma, Lin Chai, Shihan Mao, and Yucheng Zhang

***Abstract*—Source-free object detection (SFOD) aims to adapt pretrained source models to the target domain without accessing the source domain. In this article, we focus on two challenges that are unexplored in existing SFOD methods but prevalent in aerial scenes: 1) class imbalance; and 2) scale representation imbalance, i.e., the feature representations of objects at different-scales vary significantly, even within classes. To address these problems, we propose a novel SFOD method in aerial images known as dual imbalance suppression. Specifically, for challenge 1), we introduce a class-aware weighting (CAW) strategy, which adjusts the weights of each class according to the model learning state to realize dynamic balance among classes. For challenge 2), we design a scale-aware contrastive learning (SACL), which combines fine constraints on intrascale features and coarse constraints on interscale features to improve the compactness of intraclass features, while eliminating the negative impact of forcibly aligning different-scale features with significant representation differences. In addition, we propose a multiscale interaction transformer (MSIT) module that not only enhances feature representation but also constructs high-quality positive/negative sample pairs for SACL. Extensive experiments on five aerial scenes demonstrate that our method outperforms existing state-of-the-art SFOD methods.**

***Index Terms*—Aerial images, object detection, source-free domain adaptation.**

I. Introduction

DESPITE the strong performance achieved by deep convolutional network-based object detection methods [1], [2], [3], [4], [5], their success heavily relies on large-scale and high-quality labeled data. In practice, when a pretrained detector is deployed directly into a new scene, its performance suffers catastrophic degradation easily because of the domain shift between the training domain (source domain) and the testing domain (target domain). To mitigate this issue, domain adaptive object detection (DAOD) methods have been rapidly developed [6], [7], [8], [9], [10]. These methods improve the performance of the detector on the unlabeled target domain by reducing the difference between the two domains. Unfortunately, the source data may be inaccessible due to issues such as data transmission and privacy protection, thereby preventing the DAOD from working properly. This has motivated researchers to pay attention to the more challenging source-free object detection (SFOD) task, which adapts pretrained source models to the target domain without accessing the source domain.

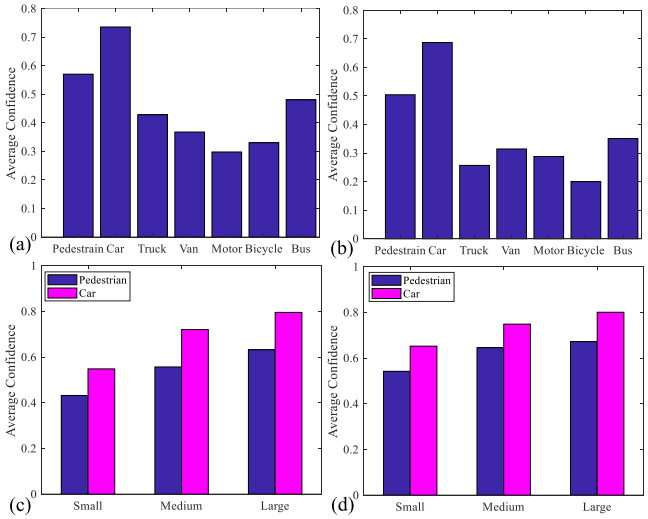


Fig. 1. Statistical analysis of proposal regions with IoU above 0.5 with Ground Truth on the cross-camera adaptation task. (a) MT model is used to evaluate the average confidence for each class. (b) Our model is used to evaluate the average confidence for each class. (c) MT model is used to evaluate the average confidence for different feature layers. (d) Our model is used to evaluate the average confidence for different feature layers. It is important to note that the MT model is only obtained using high-confidence pseudo-labels training.

Currently, existing SFOD methods usually follow the mean teacher (MT) [53] paradigm for self-training [11], [12], where the teacher model generates pseudo-labels for unlabeled target data to supervise the student model. Although these methods gain model performance improvements, they lack an effective solution to the problems of class imbalance and scale representation imbalance when dealing with aerial datasets. On the one hand, class imbalance is prevalent in aerial datasets, which discourages models from learning discriminative rare class features, further resulting in low confidence in the rare classes. However, most existing methods utilize fixed confidence thresholds to generate pseudo-labels [13], which is unfair to rare classes. In Fig. 1(a), it is clear that the average confidence of rare classes (e.g., truck and bicycle) is noticeably lower than that of large classes (e.g., person and car) when evaluated using the MT model. Since the high-confidence pseudo-labels contain fewer rare class objects, the

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rare classes are more sensitive to domain shift compared to the large classes.

On the other hand, scale variation in aerial datasets can visibly exacerbate the problem of scale representation imbalance. Even within classes, feature representations of objects at different-scales vary. In Fig. 1(c), the average confidence of specific classes on different feature layers in the teacher model is evaluated. The results show that the average confidence of the shallow layer is significantly lower than that of the middle and deep layers, which implies that there is still room for improvement in the feature representation of shallow objects (i.e., small objects). Based on this observation, the question of how to effectively enhance the feature representation to reduce the diversity of features within the class is of great importance. Considering the superiority of contrastive learning (CL) in learning high-quality representations [16], [17], it has been introduced in some SFOD methods with encouraging results [14], [15]. Nevertheless, these methods are not suitable for aerial scenarios with diverse scales, as they treat different-scale objects within the class fairly during training and force them to learn feature representations with consistency. Therefore, it is worth exploring how to embed scale information into CL.

In the task setting of SFOD, labels are not available. Due to the problems of class imbalance and scale representation imbalance, the teacher model is susceptible to missing detecting rare classes and small objects during training, so the student model is only supervised by some high-quality pseudo-labels, i.e., the student model performs biased learning. Besides, the teacher is updated as an exponential moving average (EMA) of the student model, and this pattern of collaborative learning between the teacher and student models leads to a continuous accumulation of errors, further inhibiting the performance of rare classes and small objects.

To address the above problems, we propose a dual imbalance suppression method. In terms of class imbalance, we design a class-aware weighting (CAW) strategy. This strategy adaptively generates a confidence threshold for each class, which can alleviate the issue of missing rare classes in pseudo-labels caused by high thresholds. Then, the classes are weighted according to the class thresholds to further raise the attention of the model to the rare classes. Compared to Fig. 1(a), the average confidence of each class in our method [Fig. 1(b)] is increased, especially for rare classes, such as bicycles and trucks. In terms of scale representation imbalance, we design two mechanisms: multiscale interaction transformer (MSIT) and scale-aware CL (SACL). In the former, intra and interscale feature interactions are implemented using a transformer, where intrascale utilizes self-attention for local knowledge enhancement of single-scale features, and interscale uses cross-attention for complementary knowledge integration of multiscale features. Combining two enhanced features improves the scale-awareness of the features. In the latter, interinstance fine constraints are implemented within scales to avoid direct alignment of multiscale features with obvious representation differences. Meanwhile, instance-prototype coarse constraints are applied between scales to increase the compactness of intraclass features. In particular, MSIT can provide SACL with high-quality pseudo-labels for constructing positive/negative sample pairs. Compared to Fig. 1(c), the average confidence of our method [Fig. 1(d)] is improved at different scales, especially for shallow small objects, and the average confidence difference between scales is reduced. Overall, our contributions are summarized below.

1) We innovatively propose a dual imbalance suppression method for SFOD in aerial scenes.
2) We introduce a CAW strategy that dynamically balances each class according to the learning state of the adaptive detector.
3) We design an MSIT module that enhances feature representation by fusing multiscale features, especially for small objects.
4) We propose a SACL that improves the similarity of features within classes while also eliminating the negative effect of directly aligning multiscale features with representation differences.
5) Extensive experiments on five SFOD scenes demonstrate the effectiveness and superiority of our method.

II. Related Work

A. DAOD in Aerial Images

DAOD aims to learn target detectors using labeled source domains to generalize to unlabeled target domains. Existing DAOD in aerial images can be broadly categorized into three types: feature alignment methods, pixel adaptation methods, and self-training methods. Feature alignment methods reduce the difference between the feature distributions of the source and target domains through an adversarial training manner. HANet [18] proposed a weighted multilayer feature alignment module to learn high-quality domain-invariant features. FADA [19] introduces an adversarial-based foreground alignment strategy to highlight the alignment of cross-domain foreground features, which helps to mitigate the negative transfer triggered by forcing the alignment of semantically dissimilar background features in two domains. Pixel adaptation methods reduce the shift between two domains by converting the source domain image to a similar image in the target domain. Li et al. [20] first employed a generative adversarial network (GAN) to transfer the target domain to the source-similar domain, and then joined it with the source domain for cross-domain adaptation. Shi et al. [21] constructed a generator with skip-connections based on GAN to generate a transition domain, which effectively decreases the image-level differences between the two domains. The self-training method utilizes pseudo-labels generated from the target domain as supervised information to guide model training, facilitating the asymptotic alignment of the two domains. DualDA-Net [22] embeds a dual-head co-training mechanism into the MT framework, where one of the detection heads generates high-quality pseudo-labels to supervise the other detection head. CR-Mixing [23] devised a region-selective mixing strategy, which optionally blends images from the source and target domains in accordance with pseudo-label confidence to heighten the attention of the model on foreground objects. Zou et al. [24] proposed an uncertainty-aware pseudo-label learning strategy, which generates reliable pseudo-labels for the target domain. Although these methods have achieved satisfactory performance, they require access to both the source and target domains.

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B. Source-Free Object Detection

Unlike DAOD, SFOD uses only a pretrained source model and an unlabeled target domain in the adaptation process. Currently, existing SFOD methods mainly follow the self-training paradigm, i.e., the source model is employed to generate pseudo-labels for the target samples in order to supervise the learning of the target samples. SED [25] is the initial work on this task, which adaptively generates confidence thresholds for pseudo-labels using self-entropy descent. LODS [12] learns to ignore features of the target style by combining a style augmentation algorithm with graph alignment constraints. PETS [13] alleviates the training instability issue of the MT framework by periodically exchanging the weights between the teacher and student models. Yoon et al. [26] proposed a novel low-confidence pseudo-label distillation loss, which effectively exploits false positive targets. Recently, IRG [14] and LPU [27] utilized CL strategies to increase the discriminability of features and yield satisfactory results. However, these methods model objects with a limited range of scale distributions, and thus, they treat all scale objects fairly, which is not applicable to aerial scenes. Unlike these methods, we introduce scale information into CL by implementing instance-instance fine alignment within scales and instance-prototype coarse alignment between scales, which can fully exploit the information contained in different scales. To construct high-quality positive/negative sample pairs, we use the transformer module to implement complementary enhancement of multiscale features, facilitating the generation of more accurate pseudo-labels. In addition, compared with the pseudo-label generation strategy based on fixed confidence, our method can dynamically assign confidence thresholds to each class according to the model adaptation state, which relieves inadequate learning of rare classes caused by class imbalance.

III. Methodology

A. Preliminaries

1) Problem Setting: Given a labeled source domain $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where $x_i^s \in \mathcal{X}^s$ denotes ith image, $y_i^s \in \mathcal{Y}^s$ denotes the corresponding labels (including classes and bounding boxes), and N_s denotes the number of images in the source domain. The pretrained source model Θ: $\mathcal{X}^s \to \mathcal{Y}^s$ is obtained by Faster RCNN [1] trained on the source domain. In the target domain, we have access to the dataset $D_t = \{x_i^t\}_{i=1}^{N_t}$ consisting of N_t unlabeled images, where $x_i^t \in \mathcal{X}^t$ denotes ith image. SFOD aims to adapt the pretrained source model Θ to the target domain D_t without accessing the source data D_s. In the experiments, we assume that the two domains share the same label space.

2) MT Framework: Our method is built on the MT framework, which comprises two components, namely, the teacher model Θ_{tea} and the student model Θ_{stu}. Before training starts, the two models are initialized as pretrained source models. During training, the weakly augmented target images are fed into the teacher network to obtain pseudo-labels. Subsequently, these pseudo-labels are utilized as supervision signals to direct the optimal training of the student model with the strongly augmented target image as input. The target loss L_{det} of the student model is defined as

$$L_{det} = L_{cls}^{rpn} + L_{reg}^{rpn} + L_{cls}^{roi} + L_{reg}^{roi} \tag{1}$$

where L_{cls} and L_{reg} represent classification and box regression losses, respectively. L_{cls} is computed using the cross-entropy loss function, and L_{reg} is computed by the Smooth L1 loss function. The teacher model is updated to the EMA of the student model, which is defined as

$$\Theta_{tea} \leftarrow \alpha\Theta_{tea} + (1-\alpha)\Theta_{stu} \tag{2}$$

where α is the smoothing coefficient.

B. Overall Framework

Although the MT framework is effective in distilling domain-invariant knowledge from pretrained source models, the problems of class imbalance and scale representation imbalance present in the target domain can hinder the learning of domain-specific knowledge. Accordingly, we propose a dual imbalance suppression framework. The overall framework is shown in Fig. 2. For the class imbalance problem, a CAW strategy is designed, which adaptively generates class-specific thresholds and weighting factors to increase the attention of the model on rare classes. In addition, to alleviate the scale representation imbalance, we propose two strategies, namely, MSIT and SACL. The former utilizes multiscale attention to realize complementary enhancement of different-scale features, especially for small objects. The latter treats different-scale objects nonfairly when constructing the contrastive loss, so that the information contained in different-scale features can be fully exploited.

C. Class-Aware Weighting

Class imbalance is one of the main challenges limiting the progress of aerial object detection. In Fig. 3, we show the distribution of the number of classes used for experimental evaluation in the VisDrone2021 (daytime) [28] and the DIOR [29]. It can be observed that the class distribution is extremely unbalanced. Existing SFOD methods [13] mainly utilize fixed thresholds to assign pseudo-labels to objects, ignoring the effect of class imbalance. In the adaptation process, the model is prone to being dominated by large classes, leading to inadequate learning of rare classes. This bias makes the model susceptible to generating low confidence for rare class predictions. Considering that the prediction confidence of the model for different classes will keep changing during iterative training, using a fixed threshold cannot reflect the current learning state of each class.

To address the above challenges, we propose a CAW strategy that adaptively produces class-specific thresholds for each class based on the predicted state of the model during training. Specifically, we first generate an initial class-specific threshold for each class with a Gaussian mixture model (GMM). Then, the class-specific thresholds are dynamically adjusted according to the confidence of each class during the iterative process. Finally, the class thresholds are leveraged to weight the classes in order to increase the attention of the model on rare classes. Overall, CAW consists of two main components: class-specific threshold generation and class weighting.

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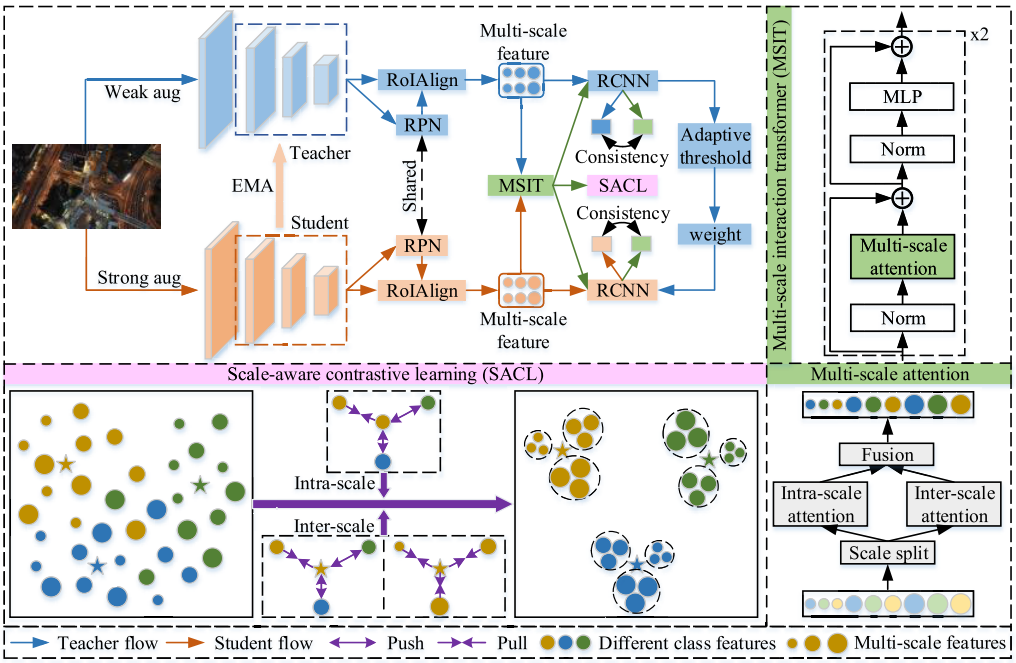


Fig. 2. Overview of our proposed framework. First, we generate region proposals from the multiscale feature layer of the teacher and student models sharing the RPN. Then, the multiscale proposal features are input to MSIT for feature enhancement, and pseudo-labels are assigned to the enhanced features using the teacher prototype. Next, the pseudo-label information is used to construct positive/negative sample pairs for SACL. Finally, CAW is applied to assist in the optimal training of the student model. Note that RCNN denotes region-based convolutional neural network; RoI denotes region of interest.

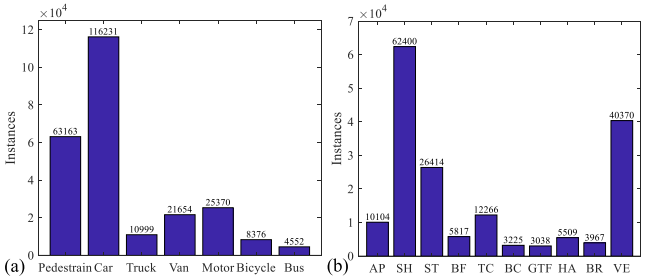


Fig. 3. Class distribution. (a) VisDrone2021 (daytime). (b) DIOR. AP: airplane, SH: ship, ST: storage tank, BF: baseball field, TC: tennis court, BC: basketball court, GTF: ground track field, HA: harbor, BR: bridge, and VE: vehicle.

1) Class-Specific Threshold Generation: As reported in the existing work [30], the fixed threshold-based method can restrict the performance of rare class objects. Therefore, we adopt a dynamic threshold updating method. Inspired by the literature [31], [32], we assume that the confidence distribution of each class follows the GMM comprising positive and negative modes. The GMM $P(c)$ of class c is defined as

$$P(c) = \sum_{m=1}^{M} \pi_m^c \mathcal{N}\left(\mu_m^c, \left(\sigma_m^c\right)^2\right) \tag{3}$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian distribution, and π_m^c, μ_m^c, and $(\sigma_m^c)^2$ denote the mixture coefficients, the mean, and the variance, respectively. $M = 2$ indicates the two modes. Subsequently, the Gaussian distribution parameters are estimated using the expectation-maximization (EM) algorithm, and the input distribution is sliced into positive and negative intervals. We select the minimum class confidence within the positive interval as the initial class-specific threshold τ_c. In methods [31], [32], they require constantly fitting a GMM model to dynamically generate the threshold during iterative training, significantly increasing the computational overhead of the model. Instead, the class-specific threshold τ_c is updated in our method to be the EMA of the class confidence means during iterative training. The class-specific threshold $\tau_c(t)$ in the tth iteration is defined as

$$\tau_c(t) = \begin{cases} \tau_c(0), & t = 0 \\ \lambda \tau_c(t-1) + (1-\lambda)\dfrac{1}{N}\displaystyle\sum_{i=1}^{N} p_i^c, & \text{otherwise} \end{cases} \tag{4}$$

where N denotes the number of RoI features, p_i^c denotes the probability that the ith RoI feature in the teacher model belongs to class c, and λ denotes the smoothing coefficient. In our experiment, $\lambda = 0.9$.

2) Class Weighting: To further improve the learning ability of the model for rare class knowledge, we reweight different classes according to class-specific thresholds. The intuition behind this strategy is that classes with low thresholds usually imply high learning difficulty and are more likely to be rare classes. Therefore, these classes need to be assigned high weights to increase the attention of the model on them. The weight w_c for the cth class is defined as

$$w_c = \begin{cases} 1, & c = \arg\max\left(\{\tau_i\}_{i=1}^{C}\right) \\ 1 + \dfrac{1 + e^{-\tau_c}}{\sum_{i=1}^{C}(1 + e^{-\tau_i})}, & \text{otherwise.} \end{cases} \tag{5}$$

We introduce the class weighting into the RoI head classification loss L_{cls}^{roi} in (1). In the tth iteration, L_{cls}^{roi} is defined

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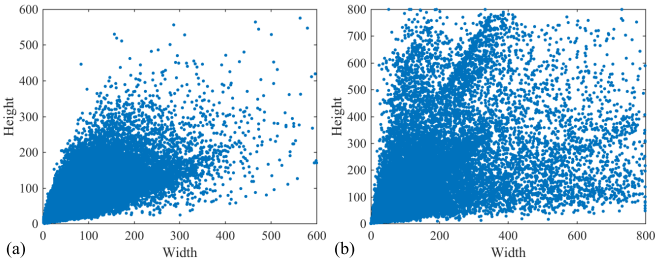


Fig. 4. Scale distribution. (a) VisDrone2021 (daytime). (b) DIOR.

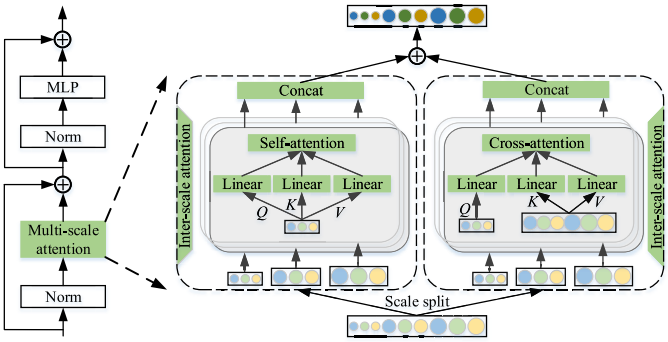


Fig. 5. MSIT module, which mainly includes intra and interscale attention.

as

$$L_{cls}^{roi} = \frac{1}{N}\sum_{i=1}^{N} w_c \cdot \mathbb{1}\left(\max\left(p_i\right) \geq \tau_{\arg\max(p_i)}\right) \cdot \mathcal{H}(\hat{p}_i, q_i) \tag{6}$$

where $\hat{p}_i$ denotes the one-hot coding of the predicted output p_i of the teacher model, q_i denotes the predicted output of the student model, $\mathcal{H}(\cdot,\cdot)$ denotes the cross-entropy loss, and $\mathbb{1}(\cdot)$ denotes the indicator function.

D. Multiscale Interaction Transformer

Although the CAW suppresses the effect of class imbalance, the severe scale representation imbalance triggered by the multiscale object distribution in the aerial dataset is also a major restricting factor for the detector performance. In Fig. 4, we show the scale distributions of the classes used for experimental evaluation in the VisDrone2021 (daytime) [28] and DIOR [29]. It is evident that these two datasets cover a wide range of object scale distributions. Existing SFOD methods mainly model datasets with a limited range of object scale distributions (e.g., Cityscapes [33] and KITTI [34]), and thus, they do not take into account the impact of scale bias on model performance. We analyze the challenges posed by scale representation imbalance in cross-domain adaptation, mainly consisting of the following two aspects. The first challenge is the obvious differences between different-scale features [35], indicating that learning a consistent representation of different-scale features directly from the same class is tricky. The second challenge is that the lack of texture detail in the small objects makes it difficult for the model to mine class-specific knowledge.

In allusion to the above challenges, we propose an MSIT module that unites intra and interscale attention to enable local knowledge enhancement of single-scale features and complementary knowledge integration of multiscale features. The MSIT module is shown in Fig. 5. Specifically, RoIs are first assigned to different feature layers according to scale size (the assignment strategy follows the literature [36]), aiming to achieve scale separation. Second, intrascale attention is leveraged to learn the dependencies between single-scale features to alleviate the first challenge. Meanwhile, interscale attention is employed to realize complementary enhancement between multiscale features to help lift the limitations of the second challenge. Finally, the outputs of the two types of attention are added together to increase the scale-awareness of the features. In the following, we describe the implementation of intra and interscale attention in detail.

1) Intrascale Attention: To tackle the first challenge in Section III-D, i.e., the inconsistent feature representations between scales, we first perform scale separation of features to model the relationships between single-scale features. Since P_2, P_3, and P_4 feature layers are used in the teacher model to generate proposals, the RoI feature layers $F^t = \{f_i^t\}_{i=1}^N (F^t \in R^{N\times d})$ in the teacher model can be separated as $F^t = \bigcup_{i=2}^4 F_i^t$ $(F_i^t \in R^{N_i\times d})$. Similarly, the RoI features $F^s \in R^{N\times d}$ in the student model that shares the region proposal network (RPN) with the teacher model, which can be separated as $F^s = \bigcup_{i=2}^4 F_i^s$ $(F_i^s \in R^{N_i\times d})$. Here, N denotes the number of RoI features, N_i denotes the number of proposals assigned in the P_i layer $(N = \sum_{i=2}^4 N_i)$, and d denotes the feature dimension. Then, for the features $F_i^{intra} = F_i^t \cup F_i^s$ $(F_i^{intra} \in R^{2N_i\times d})$ with similar scales in the teacher and student models, we obtain the three inputs of the self-attention by linear transformation, i.e., the query $Q_i^{intra} = F_i^{intra} W_{q,i}^{intra}$, key $K_i^{intra} = F_i^{intra} W_{k,i}^{intra}$ and value $V_i^{intra} = F_i^{intra} W_{v,i}^{intra}$, where Q_i^{intra}, K_i^{intra}, $V_i^{intra} \in R^{2N_i\times d}$, and $W_{q,i}^{intra}$, $W_{k,i}^{intra}$, $W_{v,i}^{intra} \in R^{d\times d}$. Next, feature enhancement is implemented using the self-attention, and the enhanced feature $\tilde{F}_i^{intra}$ is defined as

$$\tilde{F}_i^{intra} = \text{softmax}\left(\frac{Q_i^{intra}\left(K_i^{intra}\right)^T}{\sqrt{d}}\right) V_i^{intra}. \tag{7}$$

Finally, concatenating the outputs of all self-attention layers obtains the enhancement feature $\tilde{F}^{intra} \in R^{2N\times d}$ at different scales, which is defined as

$$\tilde{F}^{intra} = \text{Concat}\left(\tilde{F}_2^{intra}, \tilde{F}_3^{intra}, \tilde{F}_4^{intra}\right). \tag{8}$$

2) Interscale Attention: Since small object features contain limited information, the feature representation of small objects fails to be effectively enhanced using only intrascale attention, i.e., the second challenge in Section III-D is not adequately addressed. Previous studies have reported that multiscale feature fusion strategies can improve small object performance [37], [38], [39]. Accordingly, we propose an interscale attention to synergize with the intrascale attention to further enhance the feature representation, especially for small objects. Unlike existing feature fusion strategies, our method centers around instance-level feature interactions. Concretely, we first define the single-scale feature $F_i^{inter} = F_i^t \cup F_i^s$ and the multiscale mixed feature $F_{m_i}^{inter} = \bigcup_{j=2,j\neq i}^4 F_j^{inter} (F_{m_i}^{inter} \in R^{2(N-N_i)\times d})$. Then, the input query $Q_i^{inter} = F_i^{inter} W_{q,i}^{inter}$ $(Q_i^{inter} \in R^{2N_i\times d})$ for the cross-attention mechanism is obtained via a linear transformation of F_i^{inter}. Likewise, the key $K_i^{inter} = F_{m_i}^{inter} W_{k,i}^{inter}$ and the value $V_i^{inter} = F_{m_i}^{inter} W_{v,i}^{inter}$ are obtained by a

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linear transformation of $F_{m_i}^{inter}$, where $K_i^{inter}, V_i^{inter} \in R^{2(N-N_i)\times d}$ and $W_{q,i}^{inter}$, $W_{k,i}^{inter}$, $W_{v,i}^{inter} \in R^{d\times d}$. Next, the cross-attention mechanism is used for multiscale feature interactions and the enhancement of the feature $\tilde{F}_i^{inter}$ is defined as

$$\tilde{F}_i^{inter} = \text{softmax}\left(\frac{Q_i^{inter}\left(K_i^{inter}\right)^T}{\sqrt{d}}\right)V_i^{inter}. \tag{9}$$

Finally, concatenating the outputs of all cross-attention mechanisms obtains the multiscale complementary enhancement feature $\tilde{F}^{inter} \in R^{2N\times d}$, which is defined as

$$\tilde{F}^{inter} = \text{Concat}\left(\tilde{F}_2^{inter}, \tilde{F}_3^{inter}, \tilde{F}_4^{inter}\right). \tag{10}$$

3) Consistency Loss: The output feature $\tilde{F} = \tilde{F}^{intra} + \tilde{F}^{inter}$ of the MSIT can be further split into the teacher enhancement feature $\tilde{F}^t$ and the student enhancement feature $\tilde{F}^s$. In an effort to supervise hyperparameter learning in the MSIT, we implement this through two consistency constraints. The consistency loss L_{con} is defined as

$$L_{con} = \sum_{i=1}^{N}\left\|p_i^t - \tilde{p}_i^t\right\|_2^2 + \left\|p_i^s - \tilde{p}_i^s\right\|_2^2 \tag{11}$$

where p_i^t, $\tilde{p}_i^t$, p_i^s, and $\tilde{p}_i^s$ denote the prediction outputs of F^t, $\tilde{F}^t$, F^s, and $\tilde{F}^s$, respectively.

E. Scale-Aware Contrastive Learning

Although the MSIT strengthens the feature representation to a certain extent by learning the relationship between the features, the flaw of domain shift, which inherently restricts the representation of the original features, still exists. Currently, some SFOD methods utilize CL methods to improve the discriminability of features [14], [27], and yield satisfactory results. Inspired by this, we propose an SACL method. Unlike previous CL methods, which treat all scale targets fairly, our method embeds scale information into it. The method performs local constraints between instance pairs within scales, avoiding the direct alignment of multiscale features with obvious representation differences. Meanwhile, global constraints between instance-prototype pairs are conducted between scales to decrease the diversity of features within classes. Considering that the crucial step in constructing the contrastive loss is obtaining positive/negative sample pairs and that the RoI features are class-agnostic, we adopt a method similar to supervised CL [40], [41] by generating pseudo-labels for the features to address the above issues. It is worth noting that instead of using the pseudo-labels generated by the model directly, we opt for a prototype-based method to obtain more reliable pseudo-labels. Overall, SACL comprises two main components: prototype generation and contrastive loss.

1) Prototype Generation: Inspired by DeepCluster [42], we aggregate global features in order to attain representative prototypes that generate high-quality pseudo-labels for features. Specifically, initial prototypes are first generated using the pretrained source model, and the initial prototype K_c for class c is defined as

$$K_c = \frac{\sum_{i=1}^{N} p_{i,c}^t f_i^t}{\sum_{i=1}^{N} p_{i,c}^t} \tag{12}$$

where N denotes the number of RoI features, and $p_{i,c}^t$ indicates the probability that the ith RoI feature f_i^t belongs to class c in the teacher model. Subsequently, in order to make the most of the knowledge in the current iteration of the model, we dynamically update the global prototype using local prototypes, and the local prototype $\hat{K}_c$ for class c is defined as

$$\begin{cases}\tilde{y}_i^t = \arg\max\limits_c \dfrac{K_c \cdot \tilde{f}_i^t}{\|K_c\|_2 \cdot \left\|\tilde{f}_i^t\right\|_2} \\ \hat{K}_c = \dfrac{\sum_{i=1}^{N}\mathbb{1}\left(\tilde{y}_i^t = c\right)\tilde{f}_i^t}{\sum_{i=1}^{N}\mathbb{1}\left(\tilde{y}_i^t = c\right)}\end{cases} \tag{13}$$

where $\tilde{f}_i^t$ denotes the ith teacher enhancement feature, and $\tilde{y}_i^t$ denotes the pseudo-label of $\tilde{f}_i^t$. In (13), the teacher enhancement features with better discriminability are employed to replace the teacher features for local prototype computation. Finally, the global prototype is updated to the local prototype in the iteration by EMA, and the global prototype K_c^j during the jth iteration is defined as

$$K_c^j = \varphi K_c^{j-1} + (1-\varphi)\hat{K}_c^j \tag{14}$$

where $\varphi = 0.9$ denotes the smoothing parameter.

2) Contrastive Loss: We construct contrastive loss both intra and interscale to suppress the scale representation imbalance phenomenon. The jth student feature $f_j^s \in F_i^s (i = 2, 3, 4)$ within a single scale is selected as the anchor point, and we define the set of all intrascale features except the anchor point f_j^s as $A(j) = \{F_i^s\}\backslash\{f_j^s\}$. Also, we create the set $P(j) \equiv \{f_v^s \in F_i^s : \tilde{y}_v^s = \tilde{y}_j^s\}\backslash\{f_j^s\}$ of positive samples of f_j^s, which consists of the features with the same class as f_j^s. Since the teacher and student models share the RPN, the pseudo-labels generated from the teacher features in (13) can be taken as pseudo-labels corresponding to the student features. Intrascale CL eliminates the risk of forcing the alignment of different-scale object features with differences, whose loss L_{cl}^{intra} is defined as

$$L_{cl}^{intra} = -\sum_{F_i^s\in F^s}\sum_{f_j^s\in F_i^s}\frac{1}{|P(j)|}\sum_{p\in P(j)}\log\frac{e^{\left(\left(f_j^s\right)^T\cdot p\right)/\tau}}{\sum_{a\in A(j)}e^{\left(\left(f_j^s\right)^T\cdot a\right)/\tau}} \tag{15}$$

where τ denotes the temperature coefficient. Furthermore, interscale CL promotes the similarity of intraclass features by instance-prototype coarse alignment with a loss L_{cl}^{inter} defined as

$$L_{cl}^{inter} = -\sum_{F_i^s\in F^s}\sum_{f_j^s\in F_i^s}\log\frac{e^{\left(\left(f_j^s\right)^T\cdot k^+\right)/\tau}}{\sum_{k\in\{K_c\}_{c=1}^{C}}e^{\left(\left(f_j^s\right)^T\cdot k\right)/\tau}} \tag{16}$$

where $k^+ = K_{\tilde{y}_j^s}$ denotes the prototype with the same class as f_j^s. Finally, the scale-aware contrastive loss L_{sacl} is defined as

$$L_{sacl} = L_{cl}^{intra} + \beta L_{cl}^{inter} \tag{17}$$

where β denotes the tradeoff parameter. Scale-aware contrastive loss is dedicated to optimizing the student model to enhance the quality of student feature representations. Meanwhile, the teacher model is updated via (2). Although the SACL and MSIT superficially seem to solve the scale imbalance problem independently, they actually have a mutually reinforcing relationship. On the one hand, the MSIT enables

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enhanced feature representations and facilitates the acquisition of more accurate pseudo-labels and class prototypes. On the other hand, these accurate pseudo-labels and prototypes can be further used to construct high-quality contrastive loss, and the clustering effect of CL can, in turn, be fed back to the MSIT to generate more discriminative features.

F. Overall Objective

The total loss function of our proposed dual imbalance-suppressed framework consists of three subloss functions: the self-training detection loss L_{det} in (1) (using (6) instead of L_{cls}^{roi} in L_{det}), the consistency loss in (11), and the scale-aware contrastive loss in (17). The total loss function L is defined as

$$L = L_{det} + \gamma_1 L_{con} + \gamma_2 L_{sacl} \tag{18}$$

where γ_1 and γ_2 are two hyperparameters used to balance the loss.

IV. Experiments

A. Datasets

VisDrone2021 [28] is a challenging aerial dataset. The dataset has a total of 10 209 images divided into three subsets, i.e., a training set of 6471 images, a validation set of 548 images, and a test set of 3190 images. The dataset contains ten classes.

The UAVDT [43] consists of images selected from videos captured by UAV platforms. These data are collected from multiple scenes, including day, night, sunny, and rainy days. The dataset contains three classes, and the resolution of all the images is 1024×540. In the experiment, there are 5000 images, 1500 images, and 3500 images in the training set, validation set, and test set, respectively.

AU-AIR [44] is the first multimodal UAV dataset for object detection. The dataset is captured from different views and heights with an image resolution of 1920×1080 and covers eight classes. In our experiments, the training and test sets contain 7000 images and 3000 images, respectively.

NWPU VHR-10 [45] is composed of 800 remote sensing images, of which 715 images are derived from Google Earth, and 85 images are acquired from the Vaihingen dataset. The dataset contains ten classes: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle.

DIOR [29] is a large-scale remote sensing dataset. It is made up of 23 463 images covering 20 classes and 190 288 instances. The dataset is derived from Google Earth with a spatial resolution of $0.5 \sim 30$ m. The image size in the dataset is 800×800, and the training set and test set contain 16 424 and 7039 images, respectively.

B. Implementation Details

Following the setting of prior work [14], we adopt Faster-RCNN as the detection model, which uses ResNet50 [46] pretrained on ImageNet as the backbone network. Before training, the short edges of all images are adjusted to 600 pixels. Both source and student models are trained via a stochastic gradient descent optimizer with an initial learning rate of 0.001 and momentum of 0.9, with the batch size and epoch set to 1 and 10, respectively. In the MT framework, the smoothing coefficient α of the EMA is 0.9. To stabilize the training of the model, the teacher model is updated only once per epoch. In the experiments, the hyperparameters γ_1 and γ_2 in the overall function [see (18)] are 0.5 and 1, respectively. In the evaluation phase, we report the mean average precision (mAP) of the teacher model (i.e., Faster R-CNN) on the target domain at an IoU threshold of 0.5, and the threshold of NMS is set to 0.5. Besides, we evaluate the performance of small, medium, and large objects using mAP_S, mAP_M, and mAP_L, respectively. All experiments are implemented in PyTorch.

C. Comparison With State-of-the-Art Methods

In this section, we extensively evaluate our method on five cross-domain adaptation tasks, including cross-time, cross-camera, small-to-large-scale dataset adaptation, cross-view, and cross-weather. Compared with existing SFOD methods, our method achieves superior performance.

1) Cross-Time Adaptation: Images acquired under different time conditions can vary dramatically. The ability of the detector to work under arbitrary time periods is meaningful for its practical application. To demonstrate the effectiveness of our method on cross-time adaptation, we design two sets of experiments. One set is to select 5000 daytime images as the source domain and 1500 nighttime images as the target domain in VisDrone2021. Since the three classes, bicycle, tricycle, and awning-tricycle are almost absent in the nighttime scene, we only evaluate the performance on the other seven classes. In the other set, 5000 daytime images are selected as the source domain and 1500 nighttime images as the target domain in UAVDT.

Results on cross-time adaptation are shown in Table I. It can be seen that our method achieves optimal performance in both datasets compared to the existing SFOD method, with mAP improving by 4.1% (from 23.1% to 27.2%) and 5.0% (from 28.6% to 33.6%) over the suboptimal results, respectively. Despite the more challenging experimental settings of SFOD compared to DAOD, the results of our method are still superior to SOCCER [55]. Notably, our method shows significant performance improvement on small objects, such as "people" and "cars," which is in line with the original intention of our module design.

2) Cross-Camera Adaptation: Even in similar weather and time conditions, images captured with different equipment or parameter settings can suffer from domain shifts. To simulate this scenario, we adapt the trained source model on AU-AIR to VisDrone2021. Since the images in AU-AIR are all collected under daytime conditions, we select 5000 daytime images from VisDrone2021 as the target domain (the same as the source domain in the cross-time adaptation). Note that we rewrite the class label human in AU-AIR to pedestrian to be consistent with the class label in VisDrone2021. The performance of the adaptation detector is evaluated on seven classes shared by the two domains.

Results on cross-camera adaptation are shown in Table II. Compared to the recent SFOD methods DUR and IRG, our method obtains a performance gain of 4.8% (from 30.8% to 35.6%) and 5.3% (from 30.3% to 35.6%), respectively. Similar to the findings in Table I, the performance gains are particularly evident for the small object classes "pedestrian" and "car." Moreover, even though the class "bus" is a rare class

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TABLE I

RESULTS ON CROSS-TIME ADAPTATION. SOURCE ONLY DENOTES THAT THE PERFORMANCE OF THE PRETRAINED SOURCE MODEL IS EVALUATED DIRECTLY ON THE TARGET DOMAIN. SF DENOTES SOURCE-FREE

Methods	SF	VisDrone2021								UAVDT			
		pedestrian	people	car	van	truck	bus	motor	mAP	car	truck	bus	mAP
Source only	×	20.2	6.6	40.6	8.1	6.3	4.1	5.3	13.0	27.7	6.4	13.2	15.8
DA-Faster [47]	×	26.7	7.6	46.7	13.9	7.8	6.3	6.8	16.5	32.4	9.2	25.3	22.3
SWDA [54]	×	28.1	8.0	48.4	18.3	9.5	9.1	7.2	18.4	31.7	13.9	24.7	23.8
UMT [48]	×	29.5	10.7	52.1	22.9	13.8	14.7	9.8	21.9	35.2	19.0	29.3	27.8
AT [49]	×	30.7	10.5	52.0	19.4	13.1	16.5	12.8	22.1	33.8	18.5	30.4	27.6
HT [7]	×	31.4	11.8	51.6	19.1	12.2	18.6	**14.2**	22.7	36.1	20.6	29.3	28.7
SOCCER [55]	×	31.7	11.2	53.5	**23.9**	14.1	16.8	11.7	23.3	36.8	20.1	**31.5**	29.5
PSFOD [50]	✓	24.8	7.2	47.1	12.7	6.8	5.4	6.6	15.8	30.5	8.3	20.9	19.9
MT	✓	27.5	9.0	47.8	17.0	8.9	8.3	6.1	17.8	31.3	9.7	22.9	21.3
SOAP [51]	✓	27.3	8.2	49.3	14.7	12.2	10.6	8.8	18.7	33.6	12.3	22.5	22.8
A^2SFOD [11]	✓	30.3	10.8	51.1	20.6	14.7	15.9	9.1	21.8	35.6	19.2	25.3	26.7
IRG [14]	✓	30.4	11.8	52.6	18.2	15.3	19.6	10.4	22.4	37.1	22.9	24.2	28.1
DRU [56]	✓	31.1	12.3	53.4	19.0	15.5	20.0	10.3	23.1	37.7	23.2	25.0	28.6
Ours	✓	**35.4**	**19.7**	**60.1**	21.3	**16.7**	**24.6**	12.3	**27.2**	**43.8**	**26.8**	30.1	**33.6**

TABLE II

RESULTS ON CROSS-CAMERA ADAPTATION

Methods	SF	pedestrian	car	truck	van	motor	bicycle	bus	mAP
Source only	×	21.6	45.8	13.2	14.3	6.4	8.3	10.5	17.2
DA-Faster	×	28.7	48.0	19.6	17.1	11.8	13.7	20.3	22.7
SWDA	×	30.6	48.4	22.2	20.5	13.7	15.0	24.1	24.9
UMT	×	35.7	51.1	24.3	25.1	20.7	18.1	27.5	28.9
AT	×	32.3	53.0	27.0	28.9	19.3	18.0	31.6	30.0
HT	×	34.1	53.3	29.1	27.3	22.0	21.9	30.1	31.1
SOCCER	×	33.4	55.6	**33.0**	**31.7**	21.6	22.5	31.8	32.8
PSFOD	✓	26.1	47.7	18.7	20.1	13.4	16.2	18.3	22.9
MT	✓	30.5	50.1	19.8	22.5	15.2	17.1	20.6	25.1
SOAP	✓	27.2	50.7	21.0	25.8	13.9	16.2	25.2	25.7
A^2SFOD	✓	33.3	53.0	28.1	27.4	14.6	18.7	28.8	29.1
IRG	✓	35.2	52.6	26.7	27.2	18.1	20.3	31.9	30.3
DRU	✓	36.0	53.1	27.0	27.7	18.3	20.9	32.9	30.8
Ours	✓	**42.3**	**61.9**	30.9	29.4	**23.8**	**26.1**	**34.8**	**35.6**

TABLE III

RESULTS ON SMALL TO LARGE-SCALE DATASET ADAPTATION. AP: AIRPLANE, SH: SHIP, ST: STORAGE TANK, BF: BASEBALL FIELD, TC: TENNIS COURT, BC: BASKETBALL COURT, GTF: GROUND TRACK FIELD, HA: HARBOR, BR: BRIDGE, VE: VEHICLE

Methods	SF	AP	SH	ST	BF	TC	BC	GTF	HA	BR	VE	mAP
Source only	×	47.2	9.6	23.2	43.2	44.3	15.7	12.7	3.5	5.9	10.9	21.6
DA-Faster	×	62.8	19.9	30.4	57.2	59.8	32.3	26.6	9.5	10.6	18.7	32.8
SWDA	×	65.7	24.6	34.1	63.0	63.7	37.8	30.2	14.6	12.2	21.5	36.7
UMT	×	68.3	26.0	37.6	66.5	68.2	36.7	34.5	16.9	14.6	23.9	39.3
AT	×	70.9	27.8	39.4	68.2	67.3	38.9	33.3	18.6	14.9	23.5	40.3
HT	×	69.6	29.1	40.8	71.6	69.9	37.7	35.4	19.5	15.8	24.4	41.4
SOCCER	×	**73.5**	32.6	42.7	70.3	74.8	40.6	35.0	21.2	**17.3**	26.8	43.5
PSFOD	✓	58.2	17.6	27.8	56.4	60.1	29.6	25.3	11.5	9.2	16.6	31.2
MT	✓	65.5	20.4	36.1	64.2	66.8	37.3	28.0	13.1	10.7	21.2	36.4
SOAP	✓	64.7	22.1	34.5	66.7	69.6	36.4	30.5	13.7	10.2	20.8	36.9
A^2SFOD	✓	67.2	24.5	40.7	68.5	72.9	37.0	31.7	15.0	11.4	22.3	39.1
IRG	✓	68.5	26.3	41.3	69.2	73.2	38.9	38.5	16.6	12.2	23.2	40.8
DRU	✓	69.2	27.1	41.5	69.7	74.4	39.2	**41.5**	18.1	12.5	23.9	41.7
Ours	✓	71.5	**40.5**	**46.8**	**73.1**	**78.3**	**43.6**	40.4	**23.3**	15.6	**38.1**	**47.1**

in both domains, our method still produces good results. The benefit comes from the CAW strategy in that it improves the quality of knowledge transfer for the rare classes in the source model. The experimental results demonstrate that the design of the dual imbalance suppression strategy is reasonable.

3) Small to Large-Scale Dataset Adaptation: While it is simple to collect data in real environments, labeling large-scale datasets is time-consuming and labor-intensive. Motivated by transfer learning, we attempt to adapt knowledge from small-scale labeled data to large-scale unlabeled data. In this scenario, the small-scale NWPU VHR-10 is utilized as the source domain, and the large-scale DIOR dataset serves as the target domain. It is important to know that "baseball diamond" in NWPU VHR-10 and "baseball court" in DIOR are the same object class. For the evaluation stage, we focus only on the ten shared classes in the two domains.

Results on small to large-scale dataset adaptation are shown in Table III. Our method outperforms the state-of-the-art SFOD method DRU by 5.4% and the state-of-the-art DAOD method SOCCER by 3.6%. It is worth noting that IRG enhances the feature representation via CL in our method. The main difference is that the former ignores the interscale feature representation inconsistency problem and directly performs interinstance constraints at all scales. On the contrary, our method avoids forcing the alignment of features at different scales while also improving the compactness of the intraclass features. As shown in the table, both small object classes (e.g., VE: vehicle) and rare classes (e.g., BC: basketball court) exhibit superior results.

4) Cross-View Adaptation: VisDrone2021 and UAVDT contain images taken from multiple views, whereas DIOR only has images from top-down angles. The difference in view angles affects the representation of object information, resulting in a degradation of the detector's performance in cross-view scenes. In our experiments, we investigate the performance of our method to adapt from a single view to multiple views. Concretely, the training set in DIOR is selected as the source domain, whereas the training set in VisDrone2021 and the training set in UAVDT are used as the target domains, respectively. The detection results are evaluated only on the common class "car" of the source and target domains.

Results on cross-view adaptation are shown in Table IV. Compared to Source only, our method achieves substantial

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TABLE IV
RESULTS ON CROSS-VIEW ADAPTATION

Methods	SF	VisDrone2021	UAVDT
		AP of car	AP of car
Source only	×	22.3	17.8
DA-Faster	×	29.6	26.5
SWDA	×	33.7	35.9
UMT	×	38.3	41.5
AT	×	38.6	42.9
HT	×	39.7	43.4
SOCCER	×	41.8	44.7
PSFOD	✓	33.6	36.7
MT	✓	34.3	35.5
SOAP	✓	28.5	31.2
A^{2}SFOD	✓	37.1	41.3
IRG	✓	39.0	42.2
DRU	✓	39.4	42.9
Ours	✓	**44.2**	**46.1**

TABLE V
RESULTS ON CROSS-WEATHER ADAPTATION

Methods	SF	Rainy			Foggy			
		car	truck	mAP	car	truck	bus	mAP
Source only	×	37.4	27.6	32.5	35.2	25.5	13.0	24.6
DA-Faster	×	43.6	30.7	37.2	38.0	30.5	19.2	29.2
SWDA	×	49.5	33.2	41.4	40.8	31.6	23.5	32.0
UMT	×	51.4	34.8	43.1	41.1	33.1	27.6	33.9
AT	×	51.9	36.1	44.0	39.1	31.7	29.2	33.3
HT	×	52.6	35.7	44.2	43.5	31.9	28.6	34.7
SOCCER	×	53.3	38.1	45.7	44.9	33.8	**29.9**	36.2
PSFOD	✓	45.3	30.1	37.7	40.7	28.5	18.1	29.1
MT	✓	49.4	32.6	41.0	40.0	31.1	21.2	30.8
SOAP	✓	49.1	34.5	41.8	42.1	33.6	19.9	31.9
A^{2}SFOD	✓	51.1	35.2	43.2	43.8	34.0	24.2	34.0
IRG	✓	52.2	36.4	44.3	44.6	34.4	25.3	34.8
DRU	✓	52.8	37.4	45.1	45.0	35.2	26.2	35.5
Ours	✓	**56.5**	**40.3**	**48.4**	**51.1**	**39.7**	28.5	**39.8**

performance gains of 21.9% and 28.3% in both subtasks, respectively. On the one hand, it shows the poor generalization performance of detectors trained in single-view scenarios, i.e., applying them directly to multiview scenarios leads to a catastrophic degradation of the detector performance. On the other hand, it shows that our method manages to effectively learn multiview invariant feature representations.

5) Cross-Weather Adaptation: In the real world, the detector faces diverse weather conditions. However, since collecting images from all weather conditions is unrealistic, the detector is expected to have good robustness to weather changes. To this end, we investigate two adaptation tasks from sunny to adverse weather. In this case, 5000 sunny images are utilized as the common source domain for the two tasks, and the two target domains are 1500 rainy images and 1500 foggy images, respectively. All the above images are sourced from the UAVDT dataset. Due to the absence of the class bus in rainy conditions, only the remaining two classes (car and truck) are evaluated.

Results on cross-weather adaptation are shown in Table V. Compared with the competitive methods, our method obtained state-of-the-art performance in both subtasks, 48.4% and 39.8%, respectively. So far, extensive experiments on five cross-domain adaptation tasks demonstrate that our proposed dual imbalance suppression strategy is effective in aerial scenes.

D. Ablation Study

In order to verify the effectiveness of each module on the model, we conduct an extensive ablation study on three source-free domain adaptation tasks, including six variants: I) using the pretrained source model for performance evaluation; II) evaluating the performance of the MT model; III) adding the CAW strategy on top of variant II; IV) embedding the MSIT module into variant 2; V) introducing the SACL module into variant IV; and VI) evaluating the performance of our method. The results of the ablation study are shown in Table VI.

1) Effect of MT: Compared to the pretrained source models, the models obtained from the self-training of the MT framework exceed the mAP on the three domain adaptation tasks by 4.8%, 7.9%, and 14.7%, respectively. The remarkable performance improvement is attributed to the collaborative training between the teacher and the student in the MT framework, enabling the model to possess strong domain-invariant feature extraction capabilities.

2) Effect of CAW: The addition of CAW to variant II brings improvements in mAP of the model on all three tasks. Considering that the class imbalance problem exists in all three tasks, the design of the CAW is of practical value. However, we also find that the mAPs of the model decrease. The possible reason is that large classes in the three tasks contain a number of small objects, and CAW reduces the attention of the model to large classes, thereby making it difficult for the model to effectively detect small objects within large classes. Furthermore, through quantitative experiments, we compare the changes in the mAPs of small objects within large classes before and after the introduction of CAW. The experimental results are shown in Table VII. It is easy to observe that the mAPs for large classes decrease after the introduction of CAW.

3) Effect of MSIT: Compared to variant II, the mAP, mAP$_S$, mAP$_M$, and mAP$_L$ of the model after the introduction of MSIT are improved by 6.2%, 4.0%, 3.9%, and 5.3% on average on all tasks. The experimental results show that MSIT enhances the feature representation of objects at different scales. It should be noted that the introduction of the MSIT module increases the number of learnable parameters in the model by 1.82 M.

4) Effect of SACL: Compared to variant IV, the model in variant V obtained positive gains in all metrics on all three tasks. This indicates that the complementary enhancement between MSIT and SACL can further suppress the scale representation imbalance problem.

E. Model Analysis

1) Analysis of CAW: The results of multiple pseudo-label generation strategies on the cross-camera adaptation task are shown in Table VIII. The better performance of the class-specific threshold strategy in CAW (without using the class weighting mechanism, i.e., CAW w/o CW) compared to the fixed threshold (T) strategy suggests that there is a difference in the optimal pseudo-labeling thresholds for large and rare classes. We also notice two phenomena: 1) the performance of rare classes (e.g., bus and truck) under low

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TABLE VI
ABLATION STUDY ON THREE CROSS-DOMAIN ADAPTATION TASKS

Variants	MT	CAW	MSIT	SACL	Param. (M)	Cross-time (VisDrone2021)				Cross-camera				Small-large scale			
						mAP	mAP_S	mAP_M	mAP_L	mAP	mAP_S	mAP_M	mAP_L	mAP	mAP_S	mAP_M	mAP_L
I					28.37	13.0	5.5	19.9	31.8	17.2	8.5	24.6	34.4	21.6	11.3	25.5	41.7
II	✓				28.37	17.8	7.7	26.3	37.2	25.1	13.7	32.5	39.6	36.3	18.4	33.6	50.5
III	✓	✓			28.37	19.9	7.4	31.2	41.8	28.3	13.2	37.4	45.0	38.9	17.7	36.7	53.8
IV	✓		✓		30.19	23.1	11.3	30.1	42.5	31.7	17.4	35.6	45.5	43.1	23.1	38.3	55.2
V	✓		✓	✓	30.19	26.3	13.6	31.5	43.7	34.2	19.5	37.2	46.5	46.5	25.8	40.1	58.3
VI	✓	✓	✓	✓	30.19	27.2	13.4	33.6	45.8	35.6	19.1	40.5	49.2	47.1	25.3	41.8	59.6

TABLE VII
COMPARISON OF mAP_S ACROSS LARGE CLASSES FOR DIFFERENT METHODS ON THREE CROSS-DOMAIN ADAPTATION TASKS

Method	Cross-time (VisDrone2021)		Cross-camera		Small-large scale	
	pedestrian	car	pedestrian	car	ship	vehicle
MT	13.8	35.5	21.1	43.4	15.6	17.5
+CAW	12.7	34.9	19.8	42.5	14.1	15.9

TABLE VIII
COMPARISON OF THE PERFORMANCE AND TRAINING TIME OF DIFFERENT PSEUDO-LABEL GENERATION STRATEGIES

Methods	pedestrian	car	truck	van	motor	bicycle	bus	mAP	Time
Fixed T=0.5	28.3	47.7	20.4	20.7	13.7	17.4	21.3	24.2	2.53h
Fixed T=0.6	30.5	50.1	19.8	22.5	15.2	17.1	20.6	25.1	2.53h
Fixed T=0.7	29.9	50.3	18.8	22.0	14.7	16.6	19.4	24.5	2.53h
Fixed T=0.8	28.6	49.1	17.6	19.3	14.0	15.7	18.3	23.2	2.53h
GMM-based	32.7	52.4	23.2	25.1	16.8	18.6	23.9	27.5	2.82h
CAW w/o CW	32.7	52.1	22.5	24.6	17.4	19.2	23.4	27.4	2.57h
CAW	32.1	51.3	24.1	25.5	18.2	21.0	25.6	28.3	2.57h

TABLE IX
COMPARISON OF THE PERFORMANCE OF DIFFERENT INSTANCE FEATURE FUSION STRATEGIES. BASELINE INDICATES MODEL ADAPTATION USING THE MT FRAMEWORK

Methods	mAP	mAP_S	mAP_M	mAP_L
Baseline	25.1	13.7	32.5	39.6
+Multi-scale mixed	29.4	15.9	34.0	44.3
+Inter-scale	28.1	15.1	33.4	42.8
+Intra-scale	30.5	16.6	34.7	44.9
+MSIT	31.7	17.4	35.6	45.5

fixed thresholds surpasses that under high fixed thresholds; and 2) the performance of large classes (e.g., car and pedestrian) shows a tendency of first increasing and then decreasing as the thresholds change. The former is due to the insufficient learning of the rare class by the model, resulting in its low confidence. In the latter case, the large class contains a large number of negative pseudo-label boxes when the threshold is low, while some positive pseudo-label boxes are filtered out when the threshold is high. It is worth noting that although the previously proposed methods based on the GMM-based [31], [32] achieve a slightly higher mAP than CAW w/o CW, their training time increases more drastically. Furthermore, by introducing the class weighting mechanism, i.e., CAW, into the class-specific thresholds, the model outperforms GMM-based methods in both performance and training time.

2) Analysis of MSIT: To validate the effectiveness of intra and interscale fusion strategies in MSIT, we carry out multiple sets of experiments on the cross-camera adaptation task. The results are shown in Table IX. Compared to the baseline, after introducing either the inter or intrascale feature fusion strategy in MSIT, the model shows performance improvements in mAP_S, mAP_M, and mAP_L. Notably, when these two fusion strategies are combined, the performance of the model experiences a further improvement, implying that they can synergistically enhance feature representation. In addition, compared to the multiscale mixed feature fusion strategy, which directly inputs all features into the transformer for feature aggregation, MSIT has a significant performance advantage. This indicates that it is necessary to consider the

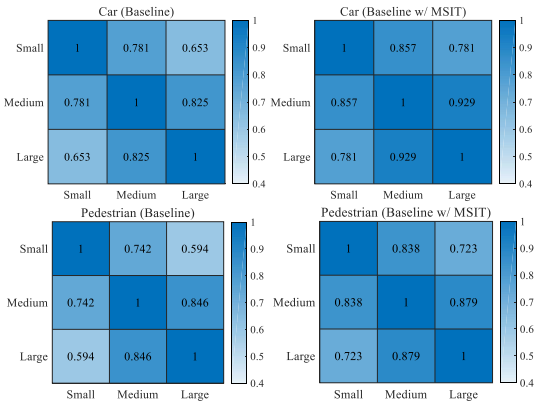


Fig. 6. Similarity between prototypes of different scales.

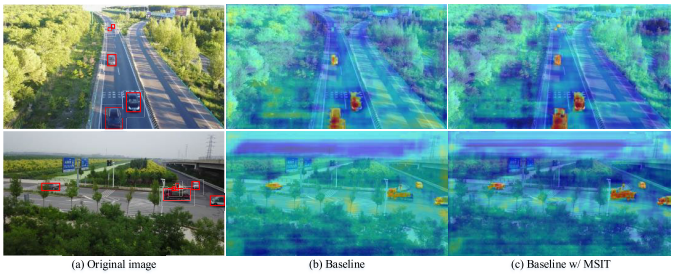


Fig. 7. Visualization of features extracted by different methods. The warmer the color, the stronger the response. (a) Original image. (b) Baseline. (c) Baseline w/ MSIT.

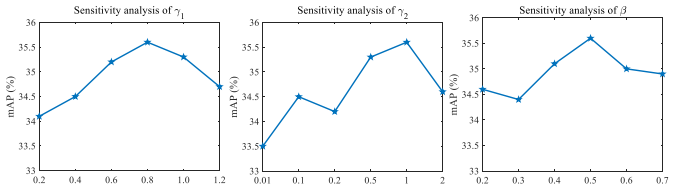
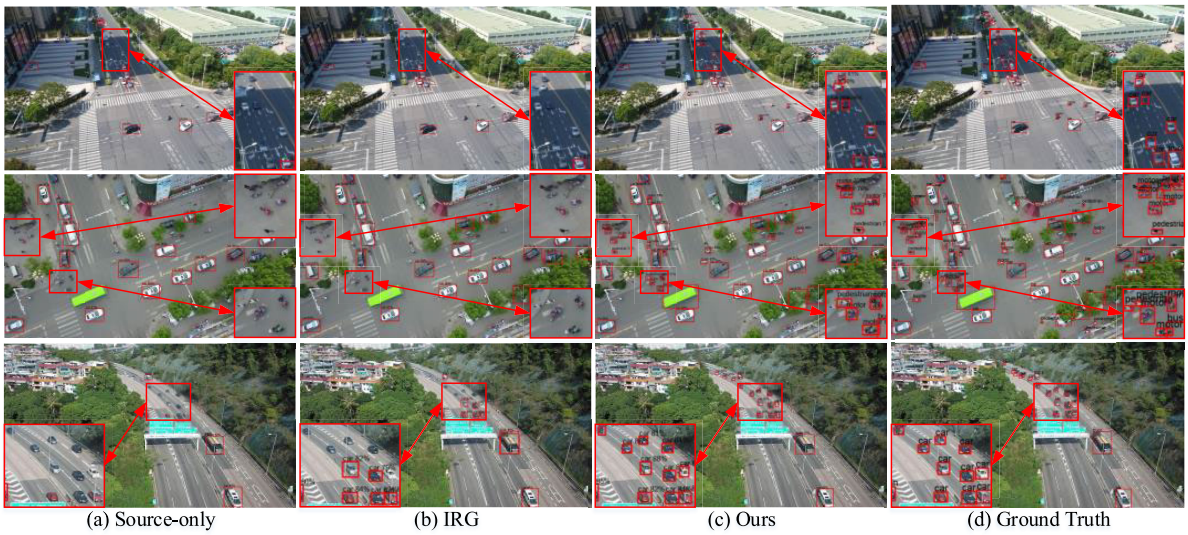


Fig. 8. Hyperparametric sensitivity analysis.

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(a) Source-only (b) IRG (c) Ours (d) Ground Truth

Fig. 9. Detection results of different methods on cross-camera adaptation. (a) Source-only, (b) IRG, (c) ours, and (d) ground truth. Zoomed-in view for the best view.

scale representation imbalance problem in the instance feature fusion process.

To further analyze the impact of the MSIT module on multiscale feature representation, we compare the changes in similarity between prototypes of different scales within the same class before and after introducing the MSIT module into the baseline. Here, a single-scale prototype is defined as the weighted average of all object features of the same class at that scale. Fig. 6 shows the similarities between prototypes of different scales for two object classes: car and pedestrian. It is evident that in the baseline, even within the same class, notable differences in similarity exist between prototypes of small and large objects, as well as medium and large ones, highlighting intraclass feature diversity. In contrast, after introducing the MSIT module, the similarity between prototypes of different scales is significantly enhanced, indicating that intraclass features are more clustered. This improvement is mainly attributed to the design of the MSIT module. It achieves local enhancement of single-scale features and global interaction of multiscale features through intra and interscale attention, effectively improving the consistency of cross-scale features.

In addition, to more intuitively demonstrate the effectiveness of the MSIT, we visualize the features extracted by the model before and after its introduction. The experimental results are shown in Fig. 7. It can be seen from the figure that the baseline has difficulty in effectively focusing on small objects. This is because the model treats objects of different scales equally, leading it to prioritize large and medium objects with rich visual information while ignoring small objects. In contrast, the model with the MSIT module responds more strongly to object regions, particularly small objects, and suppresses background regions more effectively. This further validates the rationality of the MSIT module design. Explicitly modeling the differences and relationships between different scales, it enables the model to pay balanced attention to multiscale objects, thereby improving performance.

TABLE X

COMPARISON OF THE PERFORMANCE OF DIFFERENT CONTRASTIVE LEARNING STRATEGIES. MSIT DENOTES CROSS-DOMAIN ADAPTATION USING ONLY THE MSIT MODULE. W/O INTRASCALE AND W/O INTERSCALE DENOTE WITHOUT INTRA AND INTERSCALE CONTRAST LEARNING, RESPECTIVELY

Methods	pedestrian	car	truck	van	motor	bicycle	bus	mAP
MSIT	38.6	59.3	26.4	27.3	19.7	22.2	28.4	31.7
+CL	41.1	61.6	26.9	28.0	20.5	23.2	29.7	33.0
+SACL w/o intra-scale	40.3	61.3	27.3	27.7	20.3	22.8	30.3	32.8
+SACL w/o inter-scale	41.8	62.1	27.8	28.2	20.8	23.6	30.8	33.6
+SACL	42.9	62.8	28.3	28.4	21.2	24.1	32.0	34.2

3) Analysis of SACL: The results of different CL strategies on cross-camera adaptation are shown in Table X. It can be clearly seen that adding different CL strategies to MSIT results in performance improvement of the model, implying that CL strategies can further enhance the feature representation. Compared to the traditional CL method, the mAP of our method is raised by 1.2%. In the former, interinstance constraints are directly implemented on all scale features. As we anticipated, this method neglects the differences in feature representation between scales, leading to underutilization of multiscale features. Instead, our method performs intra and interscale CL, where the intrascale aspect focuses on interinstance fine constraints and the interscale aspect concentrates on instance-prototype coarse constraints.

4) Sensitivity Analysis: We analyze the effects of the hyperparameters γ_1, γ_2 and β on the cross-camera adaptation task, where γ_1 and γ_2 are used to tradeoff the detection and adaptation losses in (18), and β is used to control the weight of the interscale contrastive loss in (17). The experimental results are shown in Fig. 8. It can be clearly seen that the hyperparameters achieve relatively stable performance over a wide range of models. In addition, the optimal parameters for γ_1, γ_2, and β are 0.8, 1, and 0.5, respectively.

5) Qualitative Detection Results: Some of the detection results on the cross-camera adaptation task, including source-

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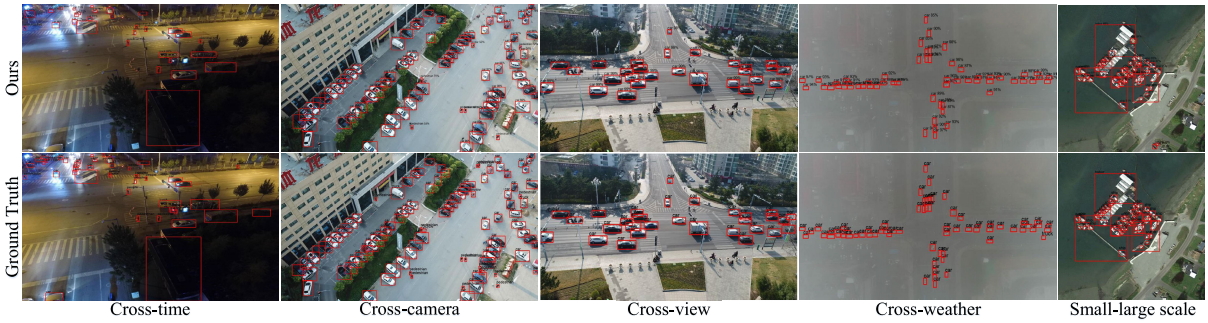


Fig. 10. Detection results on different cross-domain adaptation tasks. Zoomed-in view for the best view.

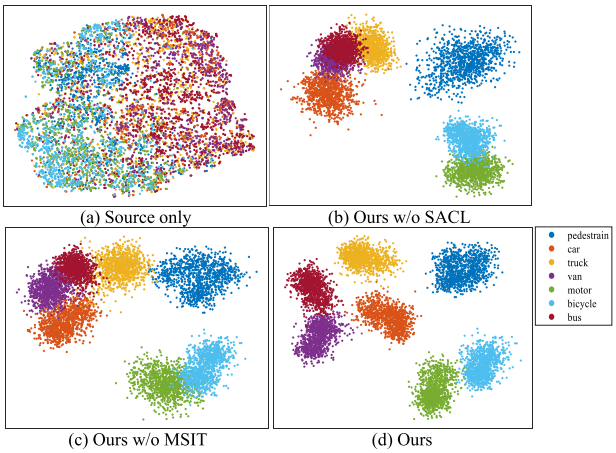


Fig. 11. Visualizing example features on the cross-camera adaptation task. (a) Directly using the pretrained source model, (b) without SACL in our method, (c) without MSIT in our method, and (d) using our method. Different colors indicate different classes. It is important to know that in (c), we directly use the labels predicted by the teachers for positive and negative sample pair construction in SACL.

only, IRG, and our method, are shown in Fig. 9. As visualized in the figure, the source-only method has poor detection performance. In comparison, IRG recognizes objects more accurately, but it fails to detect numerous small objects. Fortunately, our method effectively reduces the missed detection of small objects and realizes accurate localization of objects even in complex environments.

The detection results of our method on multiple cross-domain adaptation tasks are shown in Fig. 10. As shown in the figure, our method realizes high-quality detection of objects in conditions such as light imbalance, object occlusion, diverse views, adverse weather, and scale variability. This strongly demonstrates the feasibility of our method for cross-domain adaptation tasks in aerial scenes.

6) Feature Visualization: In Fig. 11, we visualize the features extracted by different models using t-SNE [52]. In Fig. 11(a), the features of different classes are heavily confused, i.e., the features are poorly distinguishable, indicating that there is a large shift between the source and target domains. Compared to Fig. 11(a), intraclass clustering is more evident in Fig. 11(b) and (c). This outcome is thanks to the enhancement of feature representation by means of global feature aggregation in Fig. 11(b) and the improvement of feature discriminability through the CL method in Fig. 11(c). Unfortunately, the boundaries between some similar classes in both methods are still ambiguous, e.g., between van and truck, and between motor and bicycle. In Fig. 11(d), our method can further induce intraclass compactness and interclass separation, indicating the complementarity between the two feature enhancement methods.

V. Conclusion

In this work, we focus on two key challenges commonly faced by SFOD in aerial scenes: class imbalance and scale representation imbalance. To this end, we innovatively propose a dual imbalance suppression method. Specifically, we introduce a CAW strategy, which achieves class balance by dynamically generating class-specific thresholds and weighting factors. Moreover, the multiscale interactive transformer module and SACL synergistically enhance the feature representation of objects at different scales, especially small objects, and thus effectively mitigate the scale representation imbalance. Extensive experiments on five domain adaptation tasks demonstrate that our method can achieve significant performance edges over existing SFOD methods. The method in this article provides new ideas for the application of self-training-based SFOD methods in aerial scenes.

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